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Computer Vision-Based Analysis of Volleyball Serve Trajectories

Overview

Create a tool to analyze a series of volleyball serves using computer vision and ML

- Track ball trajectory
- Predict landing zone
- Evaluate consistency



Motivation

The serve is the opening action of every volleyball rally

- A well-performed serve can decide points

Serving requires significant practice and repetition

- Tool-assisted insights can accelerate the learning process

Existing volleyball tracking tools exist, but they are:

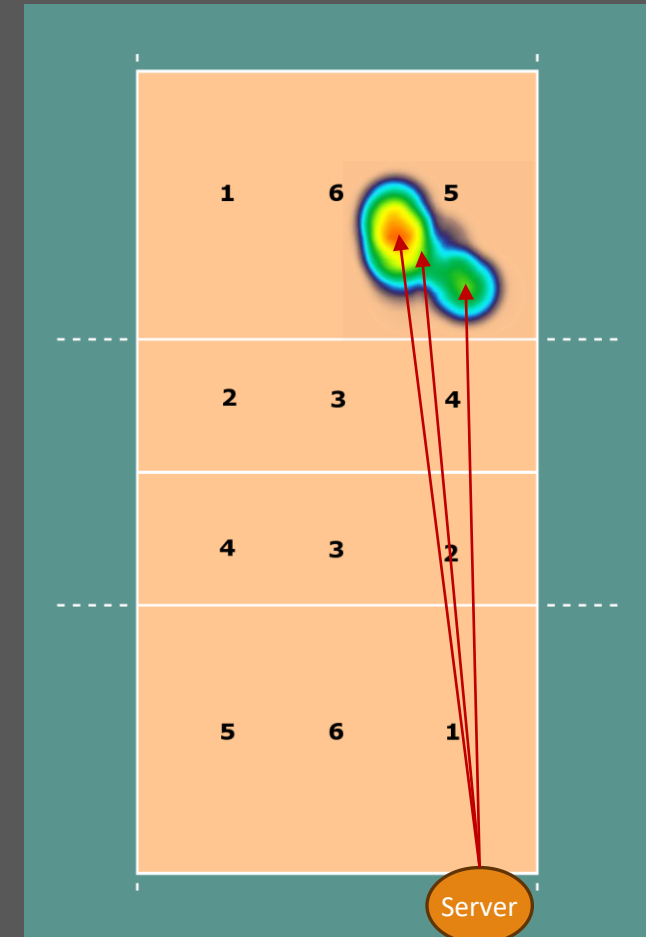
- Designed for clubs instead of individuals
- Expensive and inaccessible for most

Vision

Player records a serving practice session

Tool processes video

- Generates path visualization
- Displays landing heatmap
- Outputs relevant statistics and information



Proposed Work

Core Deliverables:

- Detect Volleyball
- Track trajectory across frames
- Determine landing zone
- Generate evaluation metrics
- Friendly UI



Future Work:

- Determine serve type (float, topspin, jump float, etc)
- Progress over time

Evaluation

Ball Tracking & Zone Classification

- Compare estimated flight path and zone vs. the actual

User Feedback

- Use surveys and testing to determine usefulness of the tool

Background

Sports CV In Other Games:

- Hawk-Eye & similar systems (tennis, cricket) provide accurate trajectories with multi-cam setups
- Costly & impractical for everyday volleyball training

Detection & Tracking Tools:

- YOLO family - fast, real-time ball detection
- PathFinderPlus – setting recognition and tracking

Current Limitations:

- Existing tools often require multi-camera setups or high-end compute

Major Milestones



LITERATURE REVIEW



DATASET COLLECTION



BALL TRACKING AND
LANDING
ESTIMATION



CONSISTENCY
TRACKING



TOOL UI



FINAL REPORT &
DEMO



Questions

